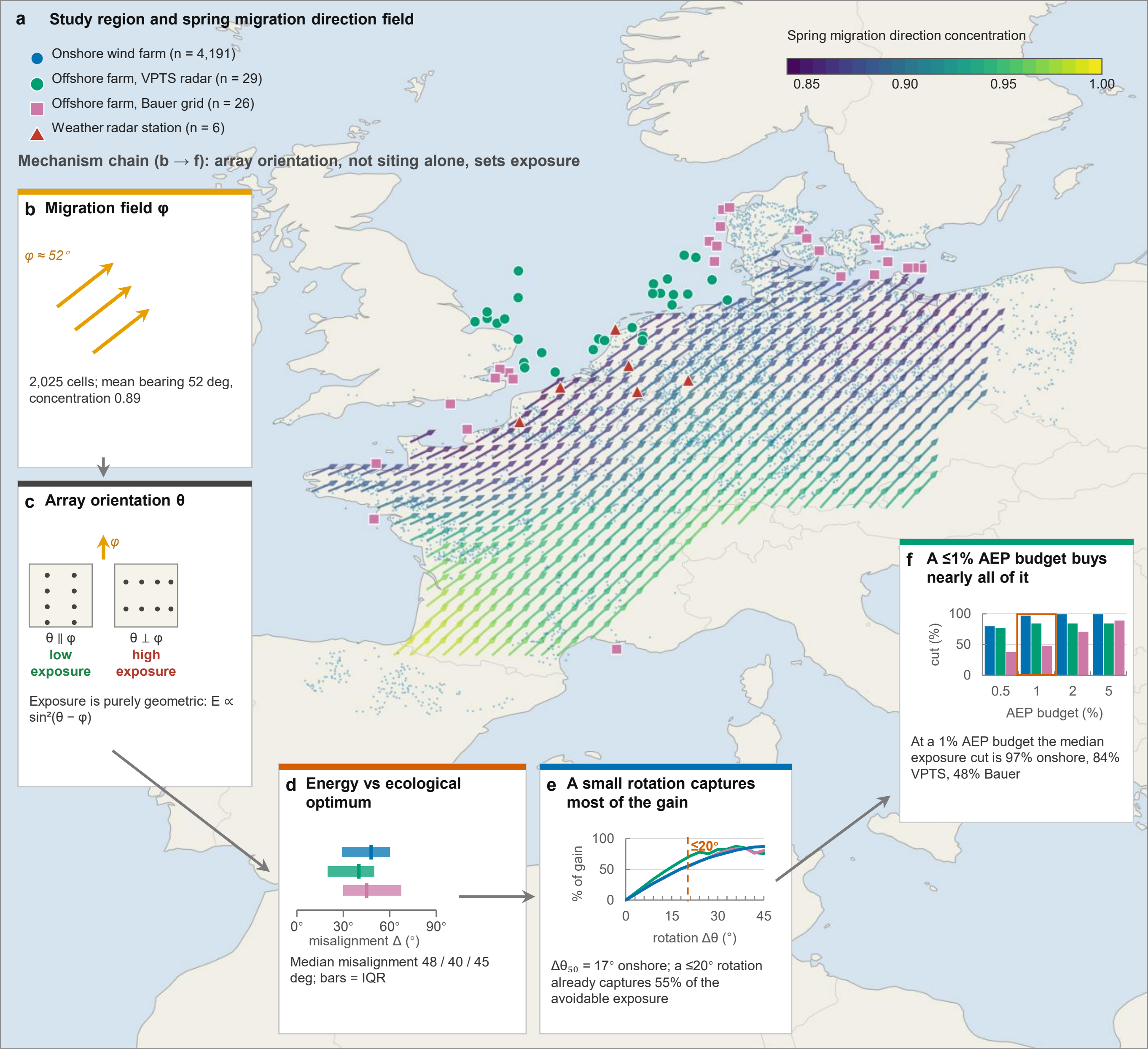

a Study region and spring migration direction field
Onshore wind farm (n = 4,191)
Offshore farm, VPTS radar (n = 29)
Offshore farm, Bauer grid (n = 26)
Weather radar station (n = 6)
Spring migration direction concentration
0.85
0.90
0.95
1.00
Mechanism chain (b → f): array orientation, not siting alone, sets exposure
b Migration field φ
φ ≈ 52°
2,025 cells; mean bearing 52 deg, concentration 0.89
c Array orientation θ
φ
θ ∥ φ
low exposure
θ ⊥ φ
high exposure
Exposure is purely geometric: E ∝ sin²(θ − φ)
d Energy vs ecological optimum
0°
30°
60°
90°
misalignment Δ (°)
Median misalignment 48 / 40 / 45 deg; bars = IQR
e A small rotation captures most of the gain
% of gain
100
50
0
≤20°
0
15
30
45
rotation Δθ (°)
Δθ50 = 17° onshore; a ≤20° rotation already captures 55% of the avoidable exposure
f A ≤1% AEP budget buys nearly all of it
cut (%)
100
50
0
0.5
1
2
5
AEP budget (%)
At a 1% AEP budget the median exposure cut is 97% onshore, 84% VPTS, 48% Bauer